7. [Maximum mark: 8]

Consider the differential equation

$$\frac{dy}{dx} = y^2 \cos^2 x \text{ , } y \neq 0$$

The solutions to this differential equation can be expressed in the form

$$y = \frac{a}{2x + \sin 2x + b} \text{ ,}$$

where a and b are constants such that $a \in \mathbb{Z}$ and $b \in \mathbb{R}$.

(a) By solving the differential equation, find the value of a. [6]

(b) Given that $y = 1$ when $x = \frac{\pi}{2}$, find the value of b. [2]